Sri Lanka Institute of Information Technology

IT3021 - Data warehousing and Business Intelligence

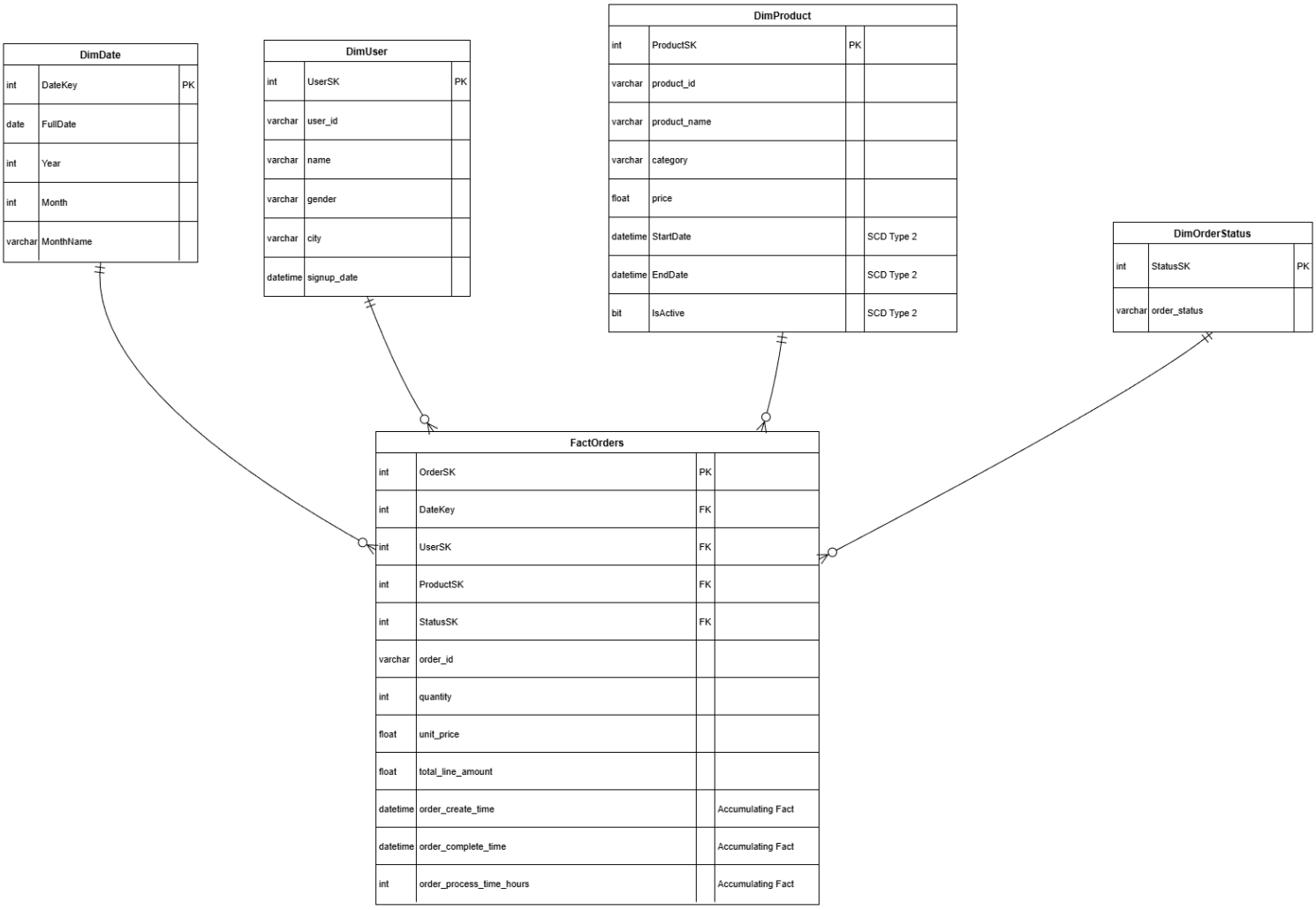
Assignment 02

IT23809024

EKANAYAKA E.J.M.C.U

Step 1: Data source for the assignment 2

Data Warehouse Design



The data source for this assignment is the Ecommerce_DW data warehouse, which was successfully designed, implemented, and loaded with data during Assignment 1. This data warehouse serves as the centralized repository for all transactional records of the e-commerce platform, specifically structured for multidimensional analysis and decision-making processes.

The warehouse follows a **Star Schema** architecture to optimize query performance and simplify data retrieval for business intelligence tasks. The schema's core is the **FactOrders** table, which captures key business metrics (measures) such as total_amount, quantity, and the calculated txn_process_time_hours. Surrounding this fact table are the following dimension tables:

- **DimUser:** Stores customer demographic details including name, email, and city.

- **DimProduct:** Contains product information such as product name, brand, category, and unit price.
- **DimOrderStatus:** Tracks the various states of an order throughout its lifecycle.

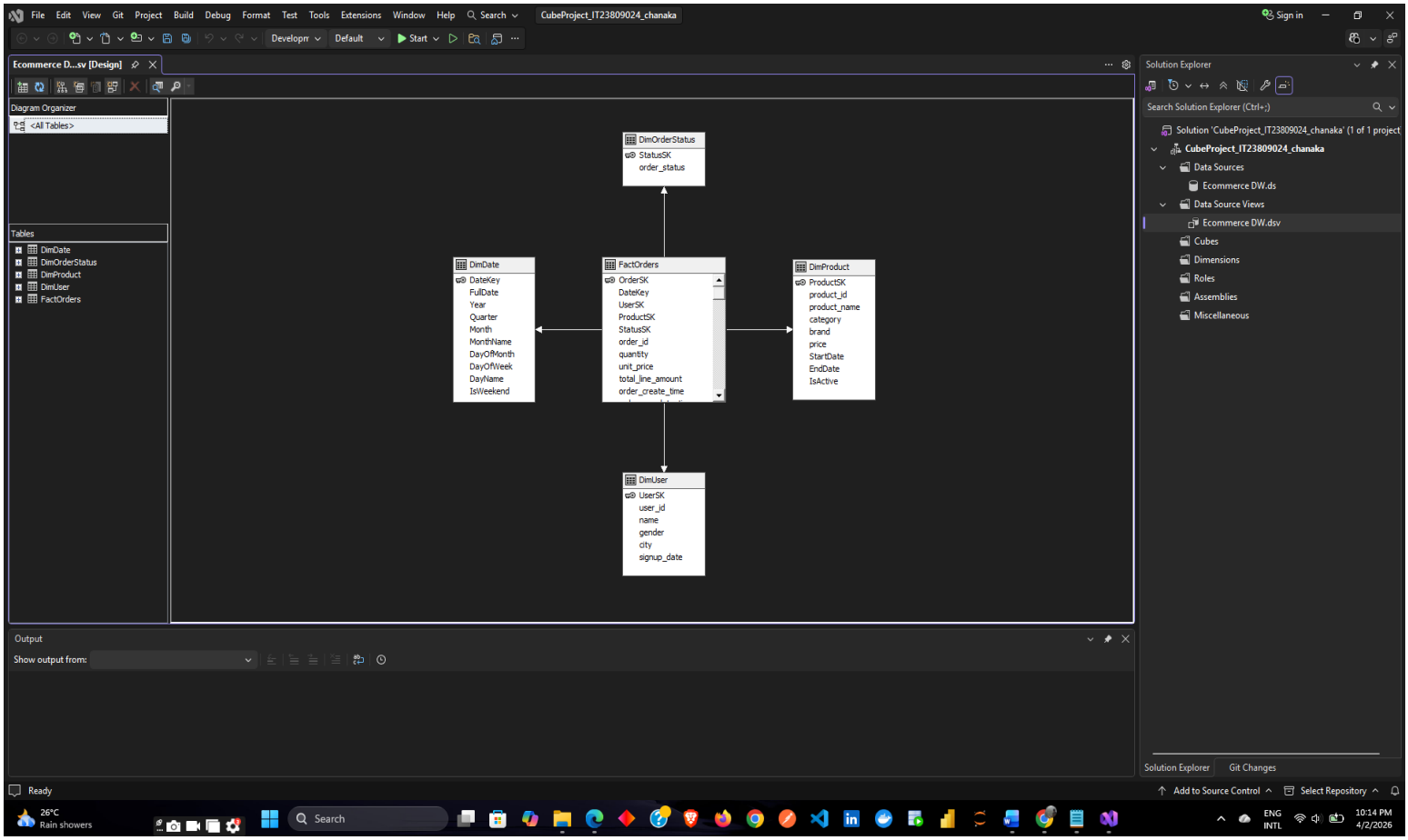
This structured environment provides a robust foundation for building the SSAS Cube and performing advanced OLAP operations in the subsequent steps of this assignment.

Step 2: SSAS Cube implementation

1. Creating a New Analysis Services Project :
The first step in constructing the OLAP cube was to set up the development environment. Using Visual Studio with SQL Server Data Tools (SSDT), a new 'Analysis Services Multidimensional and Data Mining Project' was created. The project was named 'CubeProject_IT23809024_chanaka' as per the assignment guidelines to ensure a structured approach for developing the multidimensional model.

2. Established Data Source:
A data source connection was established to link the SSAS project with the existing 'Ecommerce_DW' database. The connection was configured using the local server instance (represented by '.') and Windows Authentication. This ensures that the multidimensional model can securely access and retrieve the required fact and dimension data for cube processing.

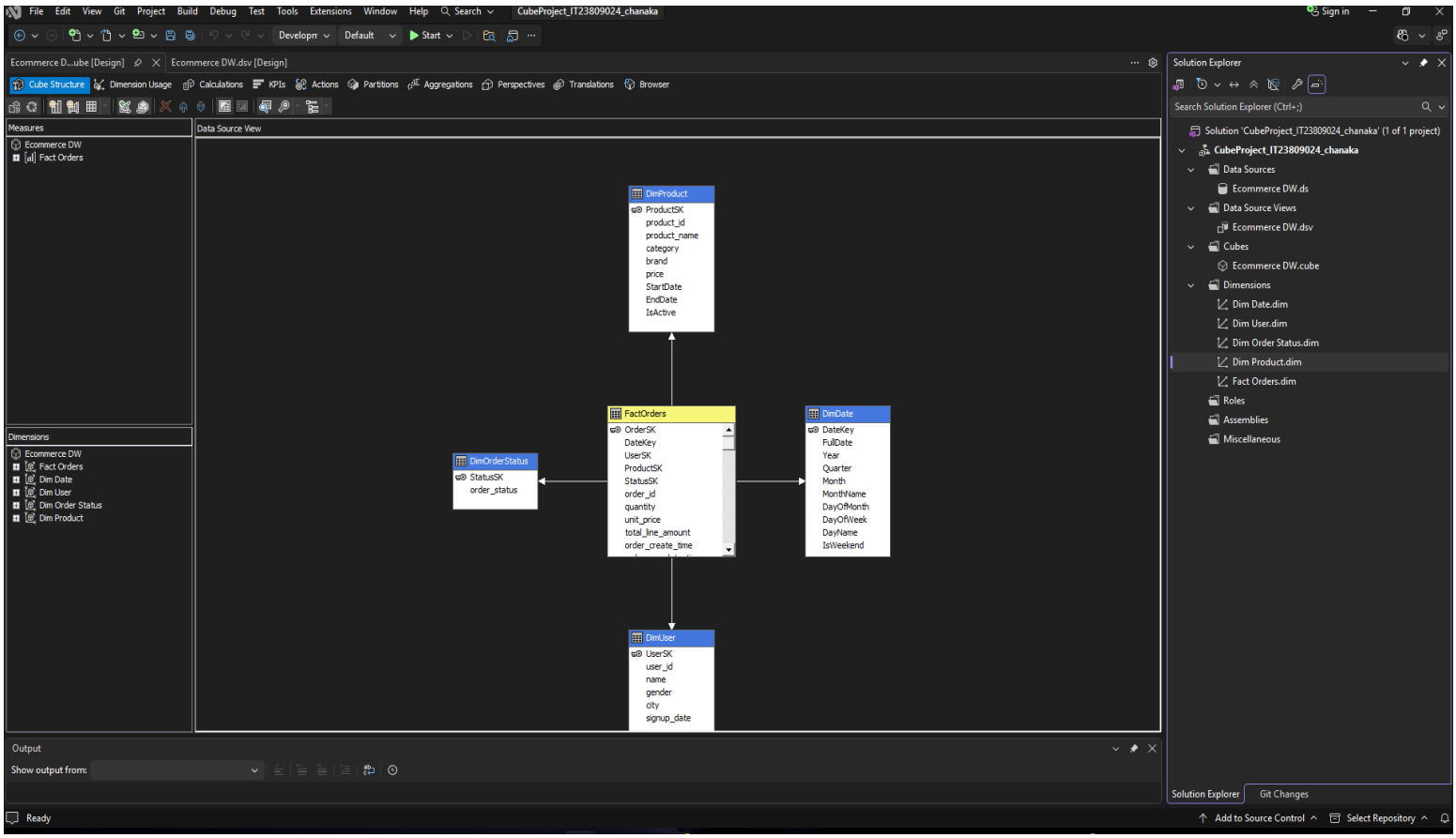
3. Building the Data Source View (DSV):
A Data Source View (DSV) was constructed to define the logical schema for the multidimensional model. The 'FactOrders' table and its associated dimensions—'DimUser', 'DimProduct', 'DimOrderStatus', and 'DimDate'—were imported from the 'Ecommerce_DW' data source. This step establishes the star schema structure, allowing for seamless integration and relationship mapping between the facts and dimensions during the cube implementation.



4. Initiated Cube Creation:
The Cube Wizard was utilized with the 'Use existing tables' option to begin the multidimensional model creation. The 'FactOrders' table was explicitly selected as the primary measure group table, as it contains the quantitative transaction data and metrics required for analytical processing and business intelligence reporting.

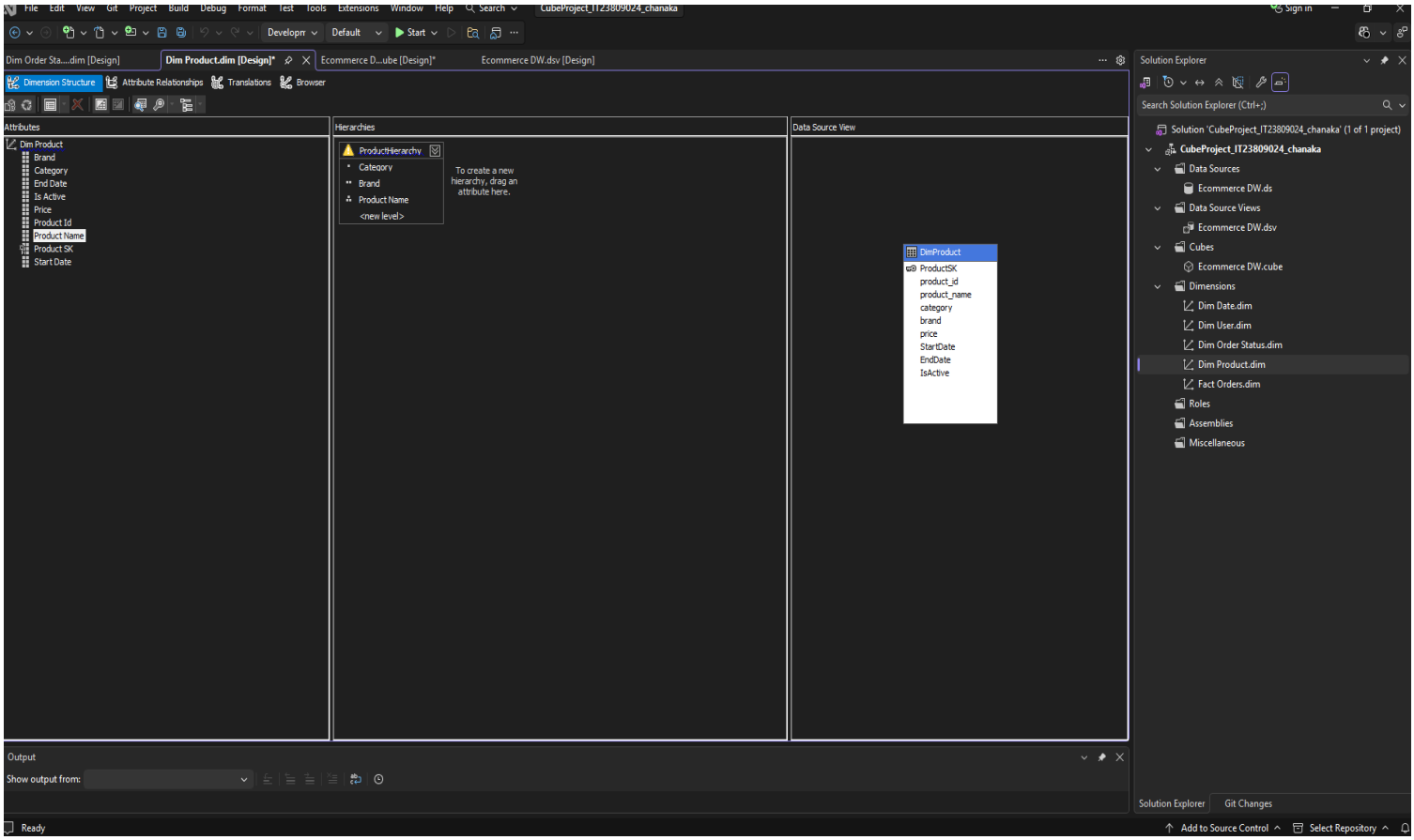
5. Selecting Measures for the Cube:
In this step, the quantitative data to be used for analysis were selected from the 'FactOrders' table. Key metrics such as 'Total Amount' and 'Quantity' were designated as measures. These measures represent the core data points that will be aggregated and calculated across different dimensions to generate business insights.

6. Incorporating Dimensions:
All relevant dimension tables were added to the cube to complete the star schema structure. This configuration includes 'DimUser', 'DimProduct', 'DimOrderStatus', and 'DimDate'. These dimensions are crucial as they allow end-users to filter, slice, and dice the quantitative data (measures) by various attributes such as customer details, product categories, order status, and specific time periods.



6. Created Hierarchies :

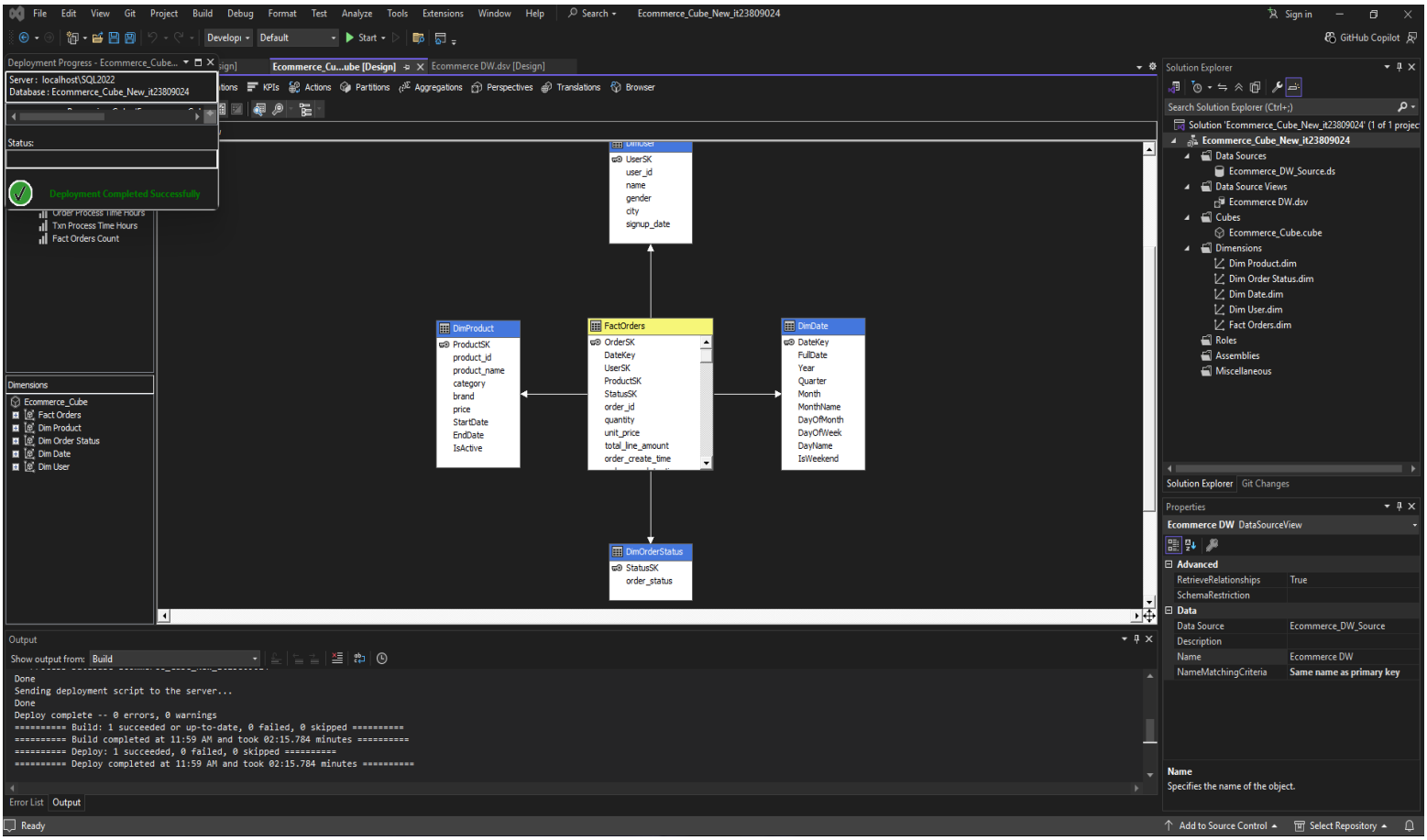
Product Hierarchy → Category – Brand – ProductName



7. Verified Relationships:
Checked and corrected dimension usage mappings between fact and dimension tables.

8. Deployed the Cube:
Configured deployment settings and successfully processed the cube to the SSAS server.

9. Validated Deployment:
Confirmed cube accessibility through both SSDT browser and SQL Server Management Studio.

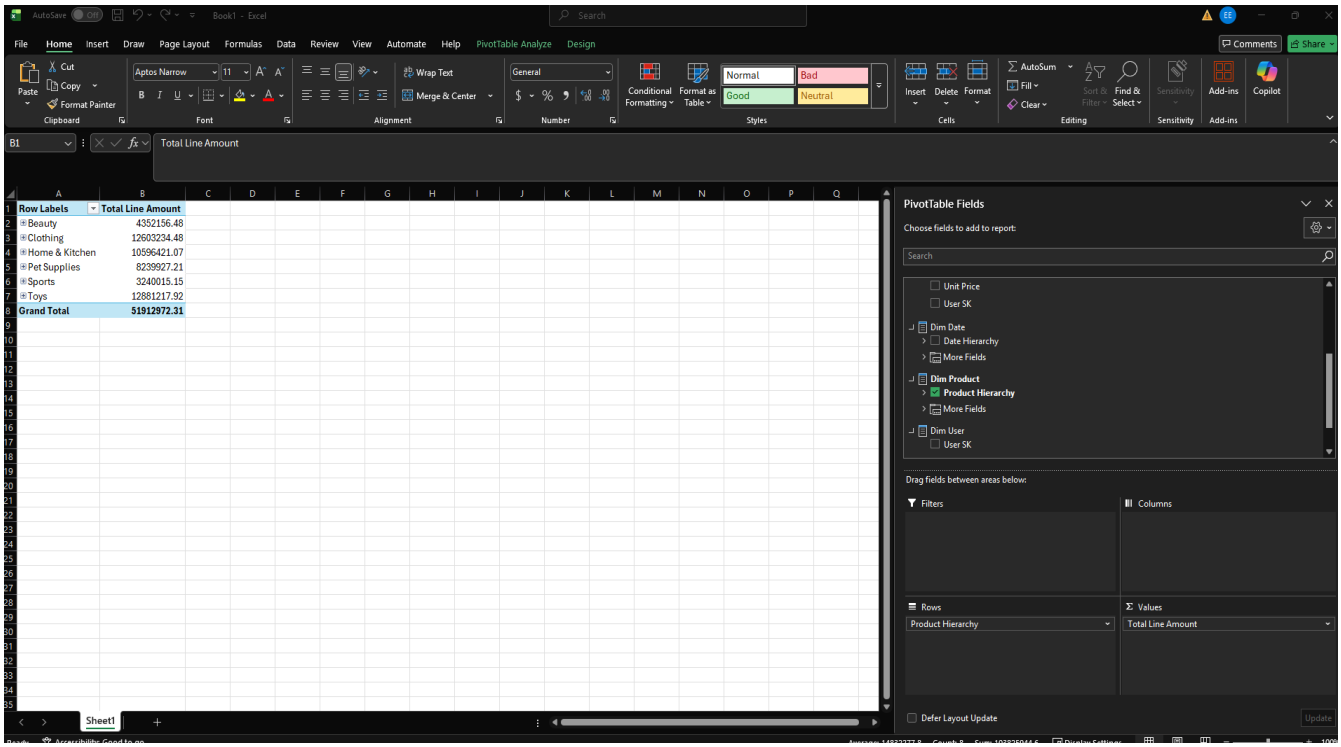


Step 3: Demonstration of OLAP operations

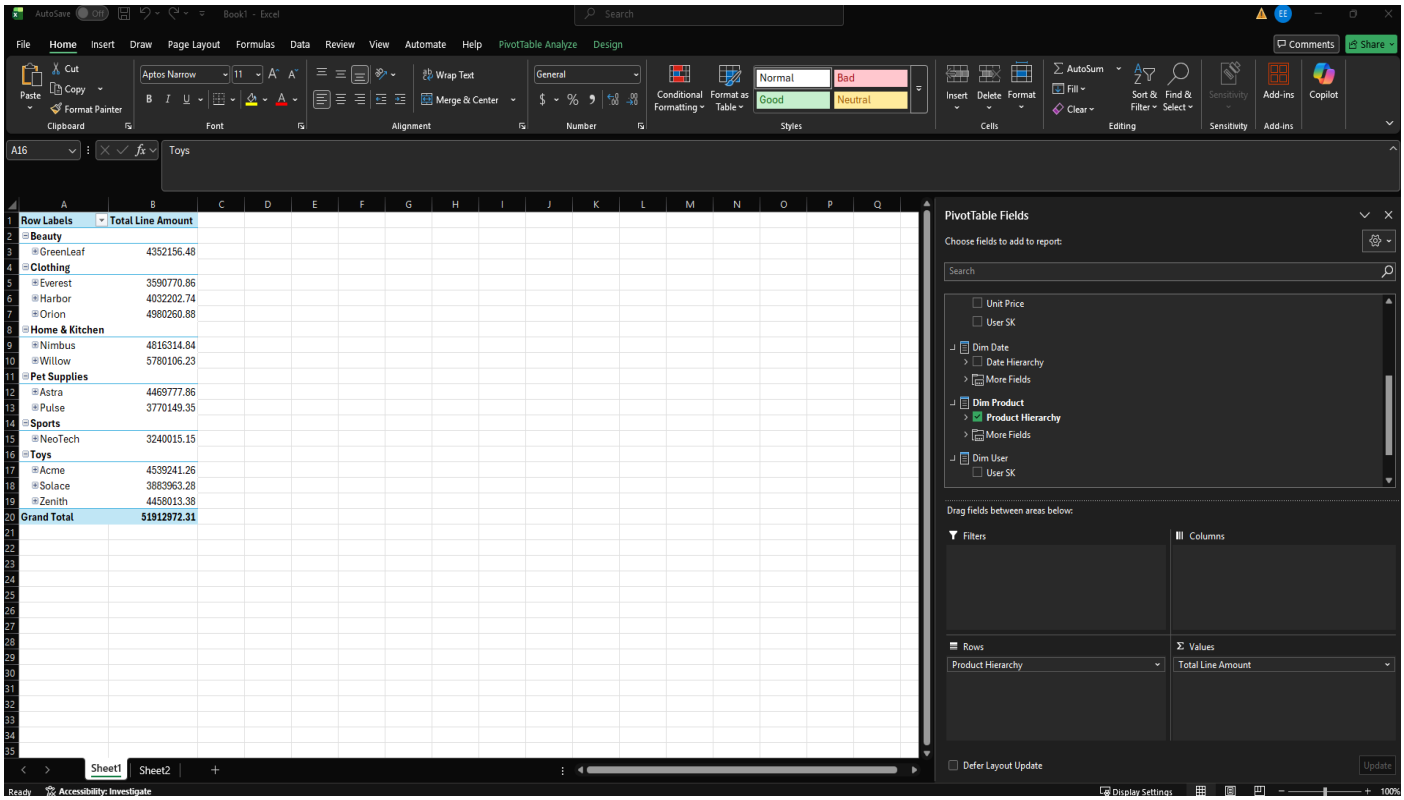
1. Connected Excel to SSAS Cube – Used Excel's Get Data feature to establish a connection to the deployed SSAS cube.

2. Created PivotTable – Generated a PivotTable report in Excel linked to the cube for OLAP analysis.

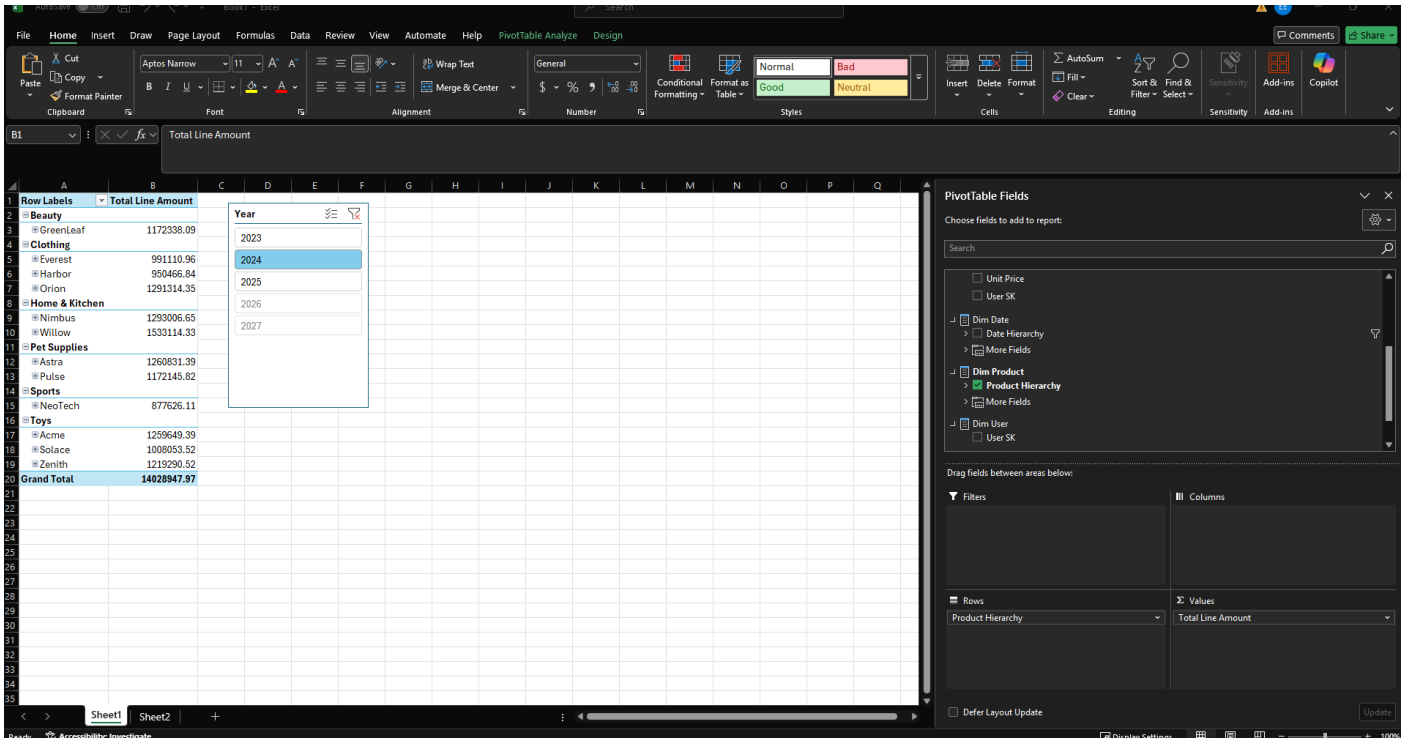
3. **Demonstrated Roll-up -** Summarizes data by moving up a hierarchy (e.g., from **Product** to **Category** level). It aggregates measures like Total Line Amount to show a higher-level summary.



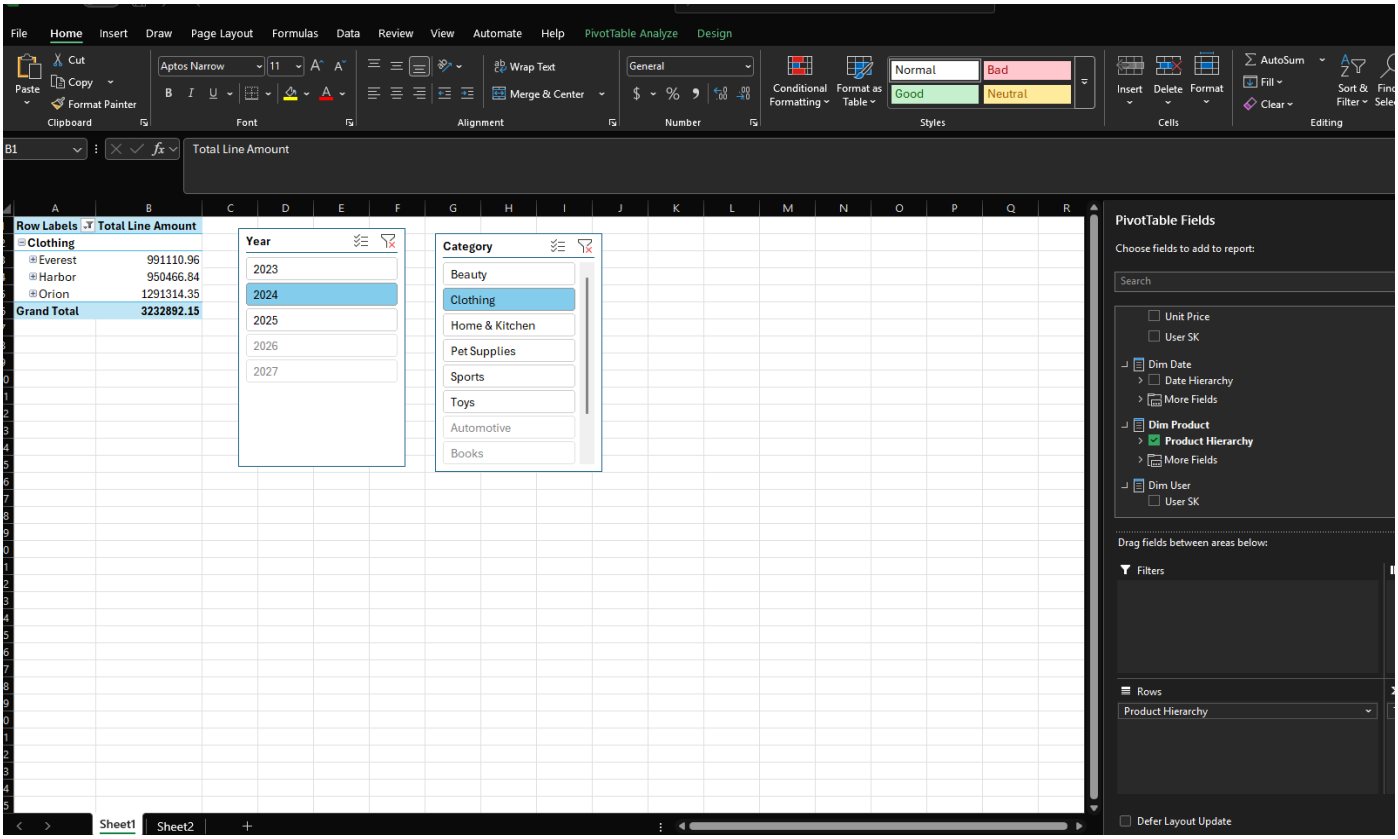
4. **Demonstrated Drill-down -** The opposite of roll-up. It breaks down summary data into more detail (e.g., clicking the "+" sign in Excel to move from Category to Brand or Product level).



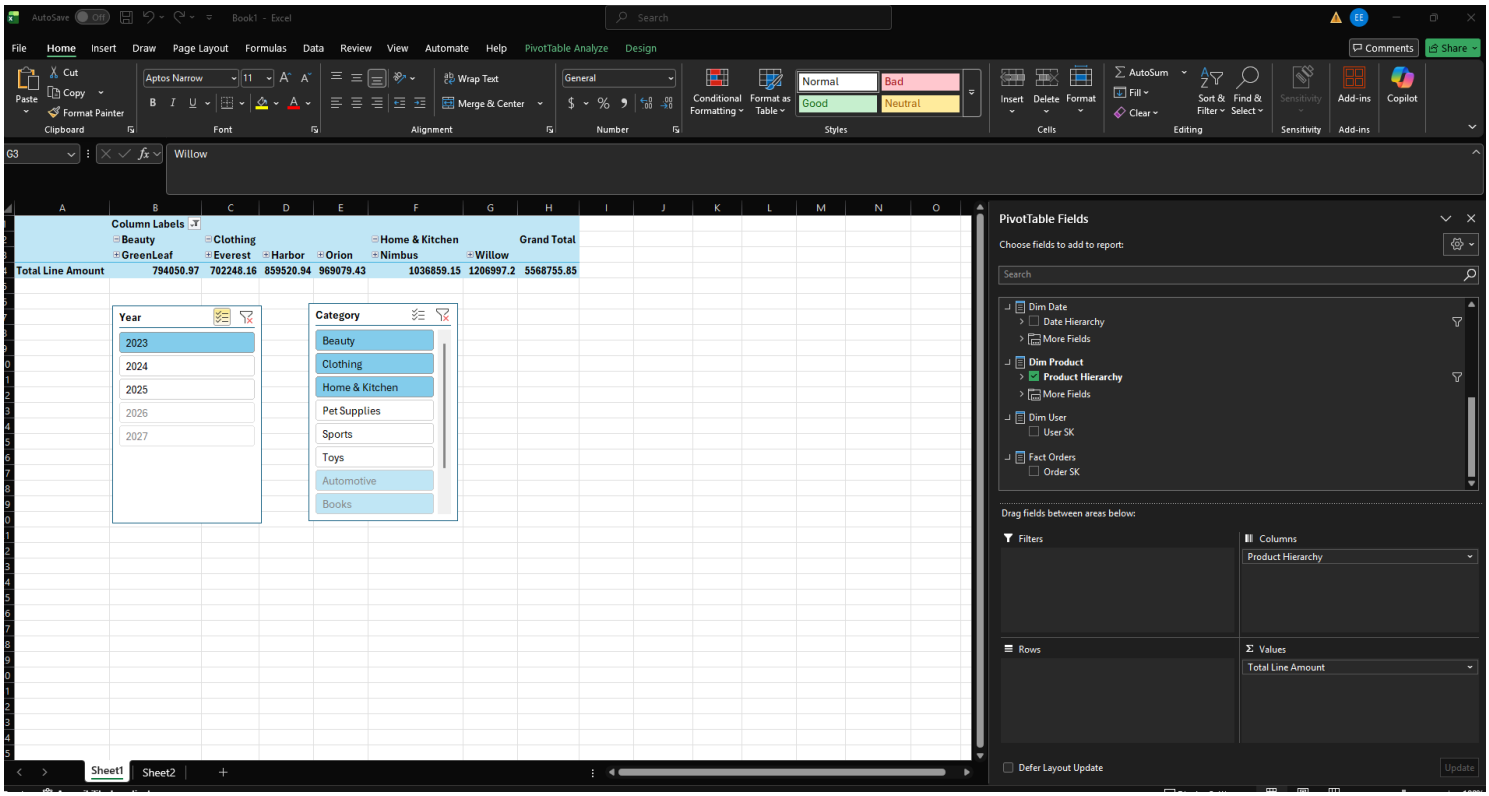
5. **Applied Slice Operation -** Filters the cube by selecting a single member from one dimension (e.g., viewing sales for only the year 2024). It creates a 2D "slice" of the data.



6. **Executed Dice Operation -** This operation selects a sub-cube from the data cube by selecting specific members from two or more dimensions. In my project, I used two Slicers simultaneously (e.g., **Year** from Dim Date and **Category** from Dim Product) to filter the data.



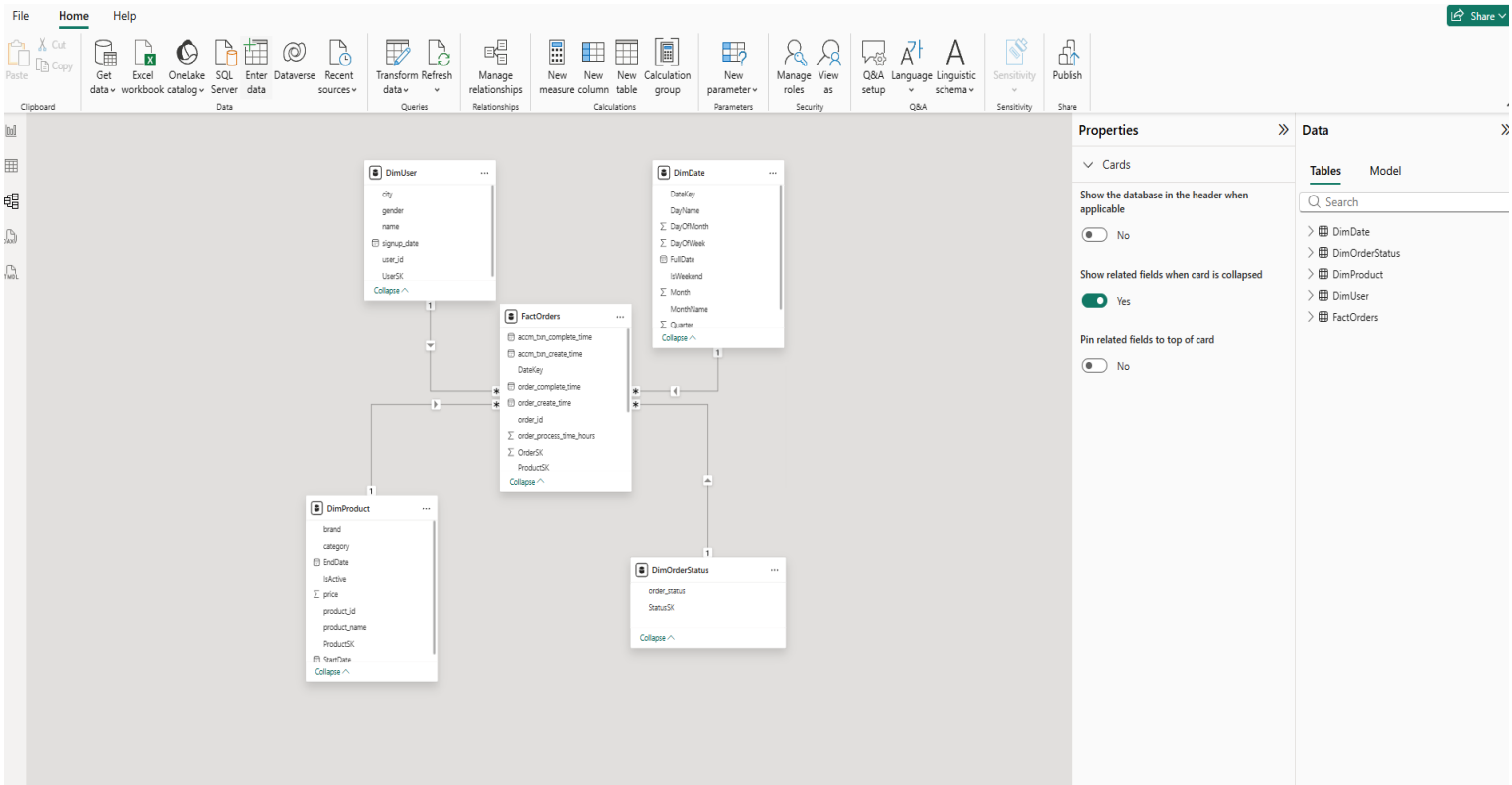
7. **Showed Pivot Functionality -** This operation changes the dimensional orientation of the data view. It involves rotating the data axes to view the data from different perspectives. In my Excel report, I moved the Category attribute from the Rows area to the Columns area to reorient the table.



Step 4: PowerBI Reports

1. Connected to Data Source – Imported fact and dimension tables from the SQL Server Analysis Services Cube into Power BI Desktop.

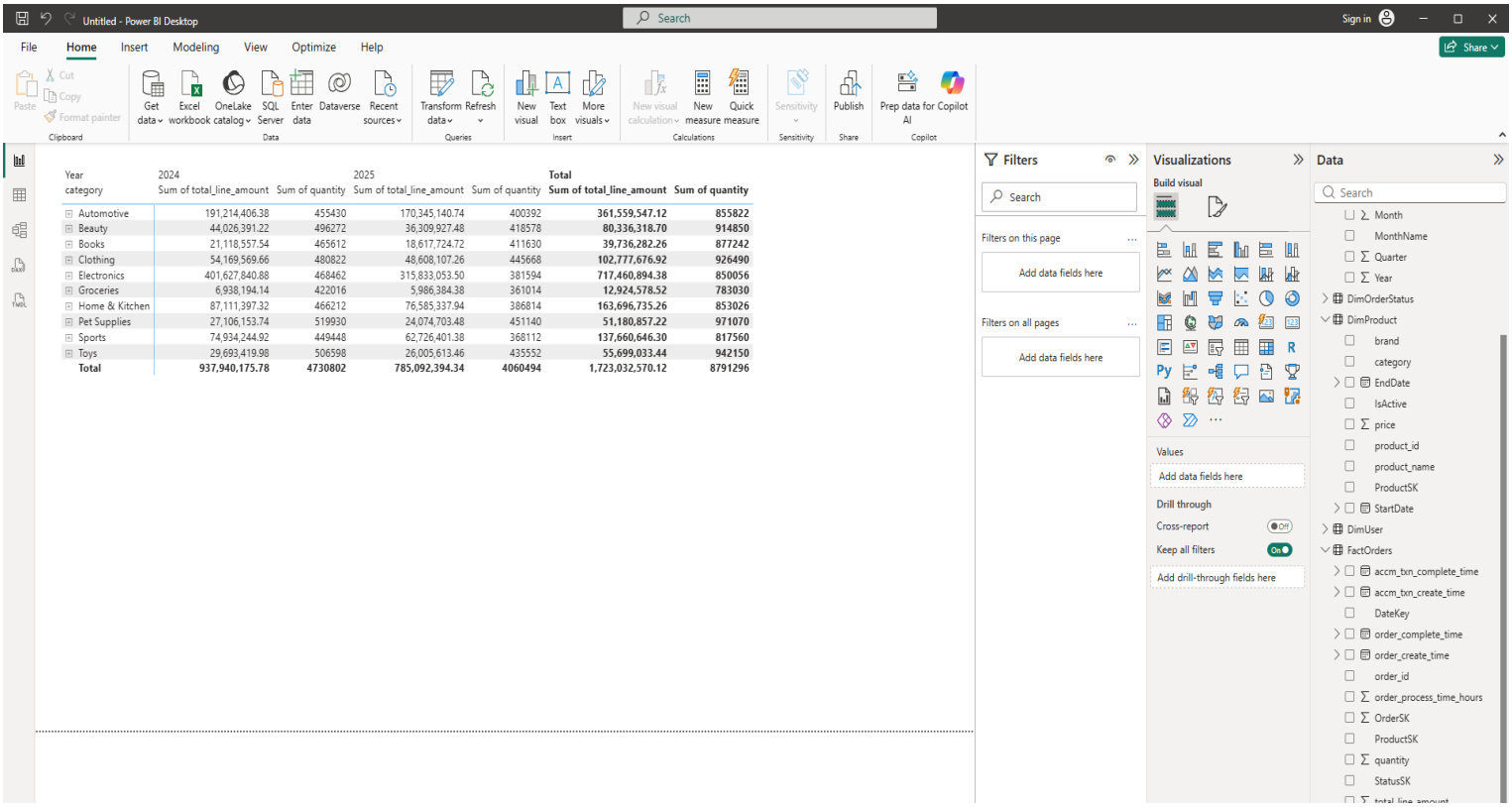
2. The star schema was already linked with the fact table to the dimension tables using key columns since we uploaded the cube in the last step.



3. Report 1: Matrix Visual

a) Designed Matrix Layout:

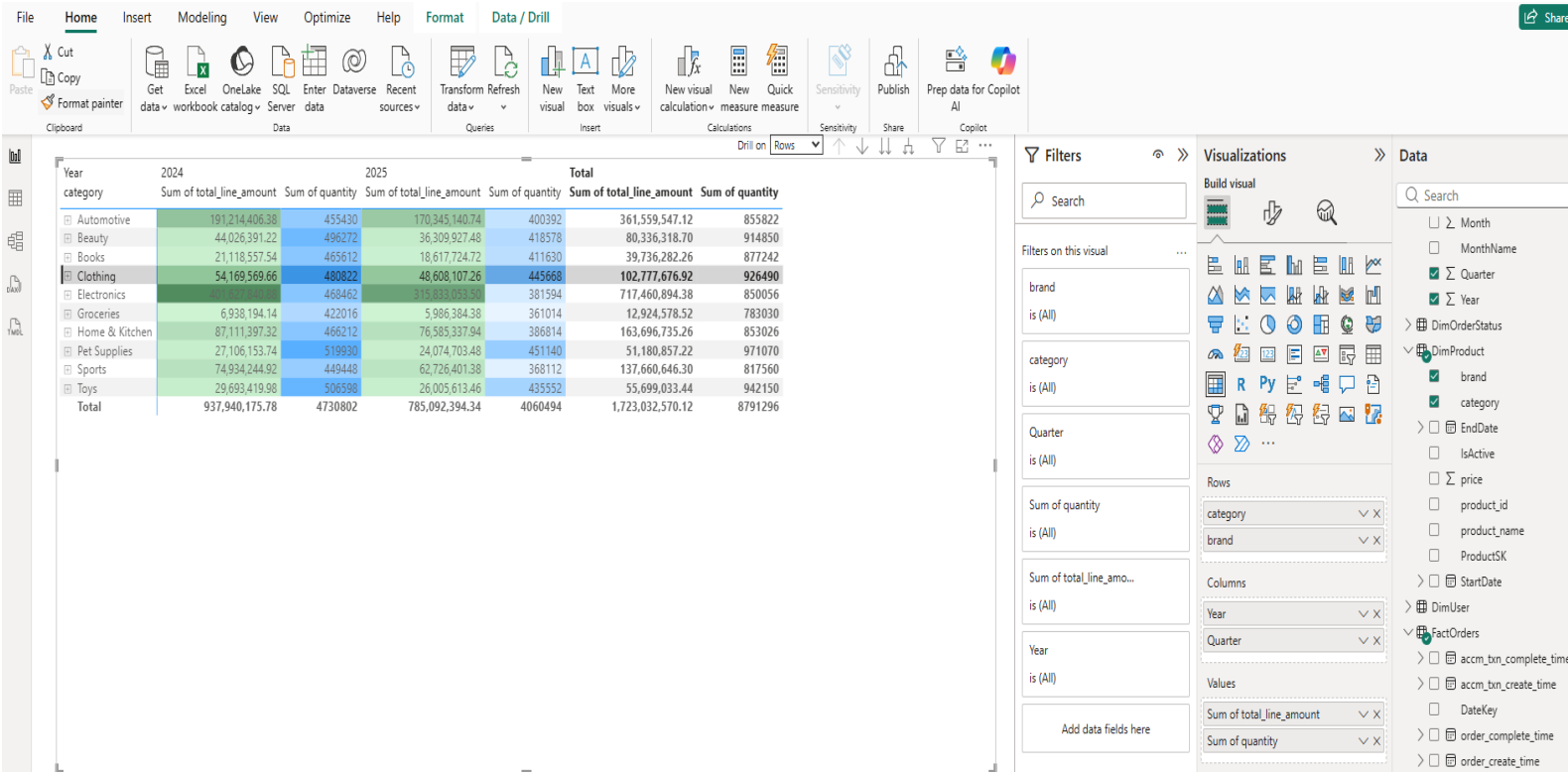
- Added Hierarchical Rows → Category, Brand
- Added Columns→ Year, Quarter
- Added Values → Total Line Amount, Order Quantity



Year	2024		2025		Total	
category	Sum of total_line_amount	Sum of quantity	Sum of total_line_amount	Sum of quantity	**Sum of total_line_amount**	**Sum of quantity**
Automotive	191,214,406.38	455430	170,345,140.74	400392	**361,559,547.12**	**855822**
Beauty	44,026,391.22	496272	36,309,927.48	418578	**80,336,318.70**	**914850**
Books	21,118,557.54	465612	18,617,724.72	411630	**39,736,282.26**	**877242**
Clothing	54,169,569.66	480822	48,608,107.26	445668	**102,777,676.92**	**926490**
Electronics	401,627,840.88	468462	315,833,053.50	381594	**717,460,894.38**	**850056**
Groceries	6,938,194.14	422016	5,986,384.38	361014	**12,924,578.52**	**783030**
Home & Kitchen	87,111,397.32	466212	76,585,337.94	386814	**163,696,735.26**	**853026**
Pet Supplies	27,106,153.74	519930	24,074,703.48	451140	**51,180,857.22**	**971070**
Sports	74,934,244.92	449448	62,726,401.38	368112	**137,660,646.30**	**817560**
Toys	29,693,419.98	506598	26,005,613.46	435552	**55,699,033.44**	**942150**
Total	**937,940,175.78**	**4730802**	**785,092,394.34**	**4060494**	**1,723,032,570.12**	**8791296**

b) Configure the Matrix:

- **Applied Conditional Formatting:** Implemented Background Color scales for the **Total Line Amount** (gradient from light green to dark green) and **Order Quantity** (gradient from light blue to dark blue) to highlight high-performing categories and products.
- **Subtotals and Totals:** Enabled row and column grand totals to provide a comprehensive summary of sales performance across all dimensions.
- **Visual Adjustments:** Adjusted column widths, text alignment, and font styles to improve the readability and overall aesthetic of the tabular data.
- **Data Preparation:** Used explicit measures for calculations to ensure the Matrix visual dynamically updates based on the selected time periods and product levels.



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Total	937,940,175.78	4730802	785,092,394.34	4060494	1,723,032,570.12	8791296

4. Report 2: Multiple Slicers with Cascading Filters

a) Created Interactive Filters:

- **Cascading Slicers:** Added slicers for **Category** and **Brand** from the `DimProduct` table.

- **Dynamic Interaction:** Configured the report so that selecting a specific **Category** dynamically filters the available options in the **Brand** slicer, ensuring a focused data exploration experience.

b) Added Supporting Visuals:

- **Bar Chart:** Visualized `Total Sales by Brand` to identify top-performing brands within selected categories.

- **Line Chart:** Displayed the `Monthly Sales Trend` to analyze seasonal patterns and revenue fluctuations over time.

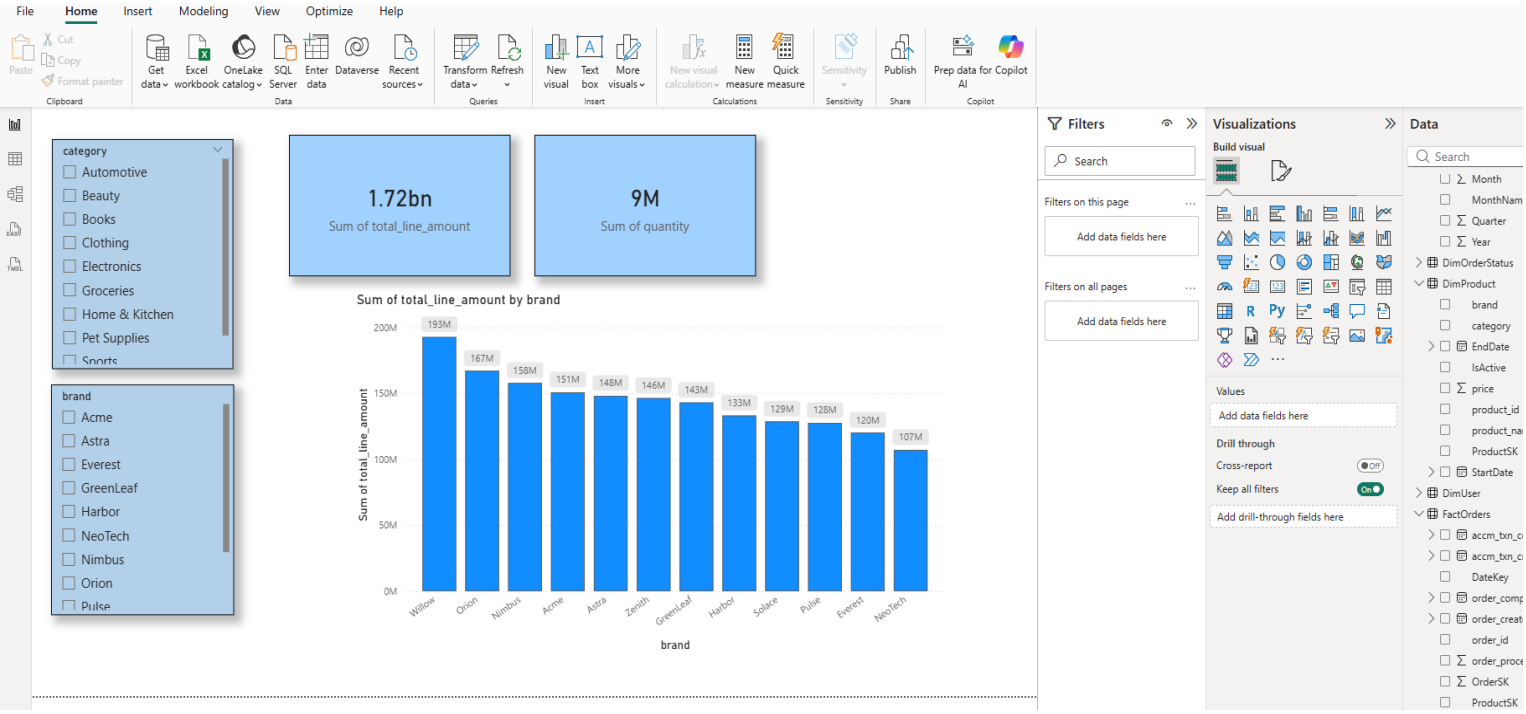
- **Card Visuals:** Created summary cards for `Total Sales Revenue` and `Total Quantity Sold` to provide instant high-level insights.

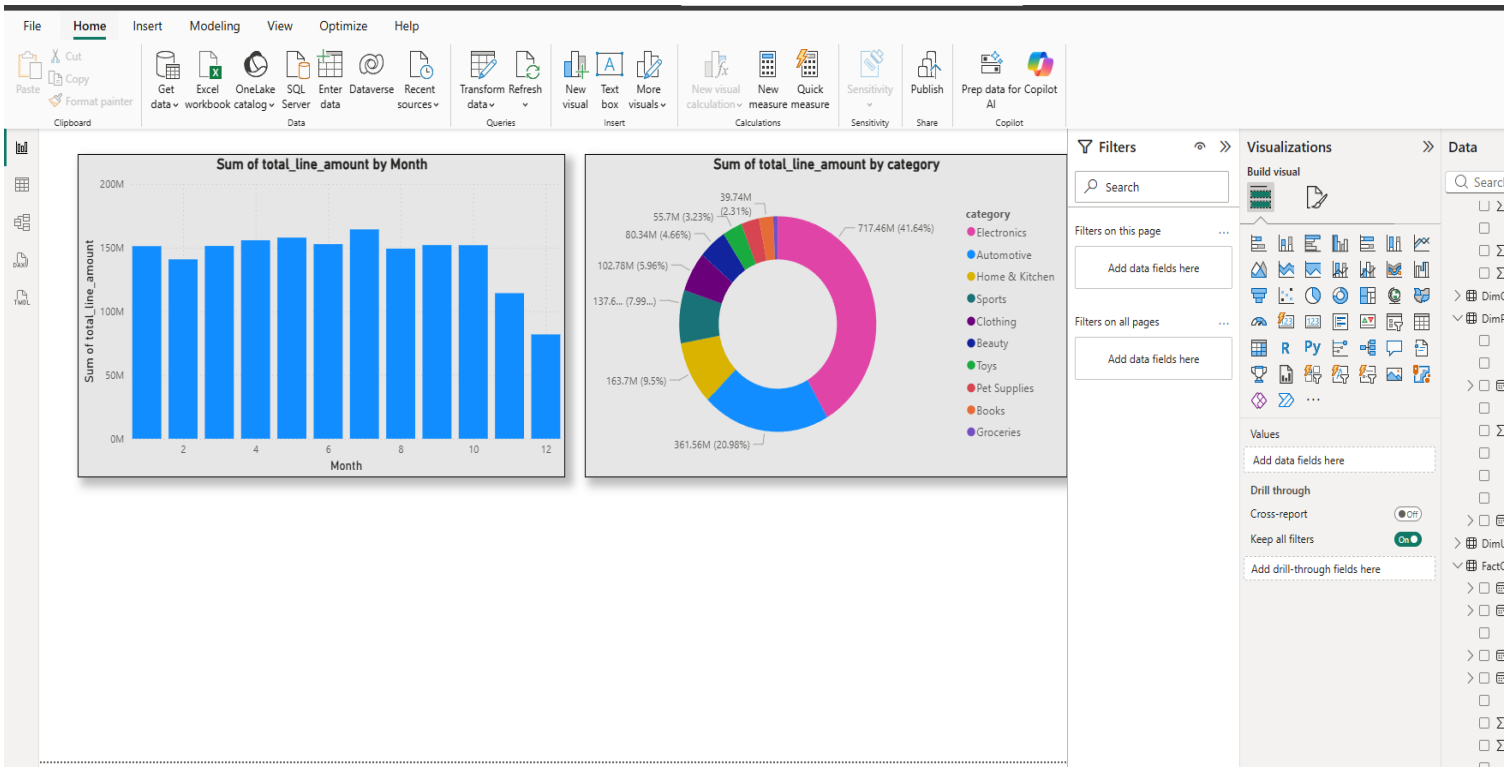
- **Donut Chart:** Showcased the `Sales Distribution by Category` to understand the contribution of each segment to the total revenue.

c) Insights & DAX Usages:

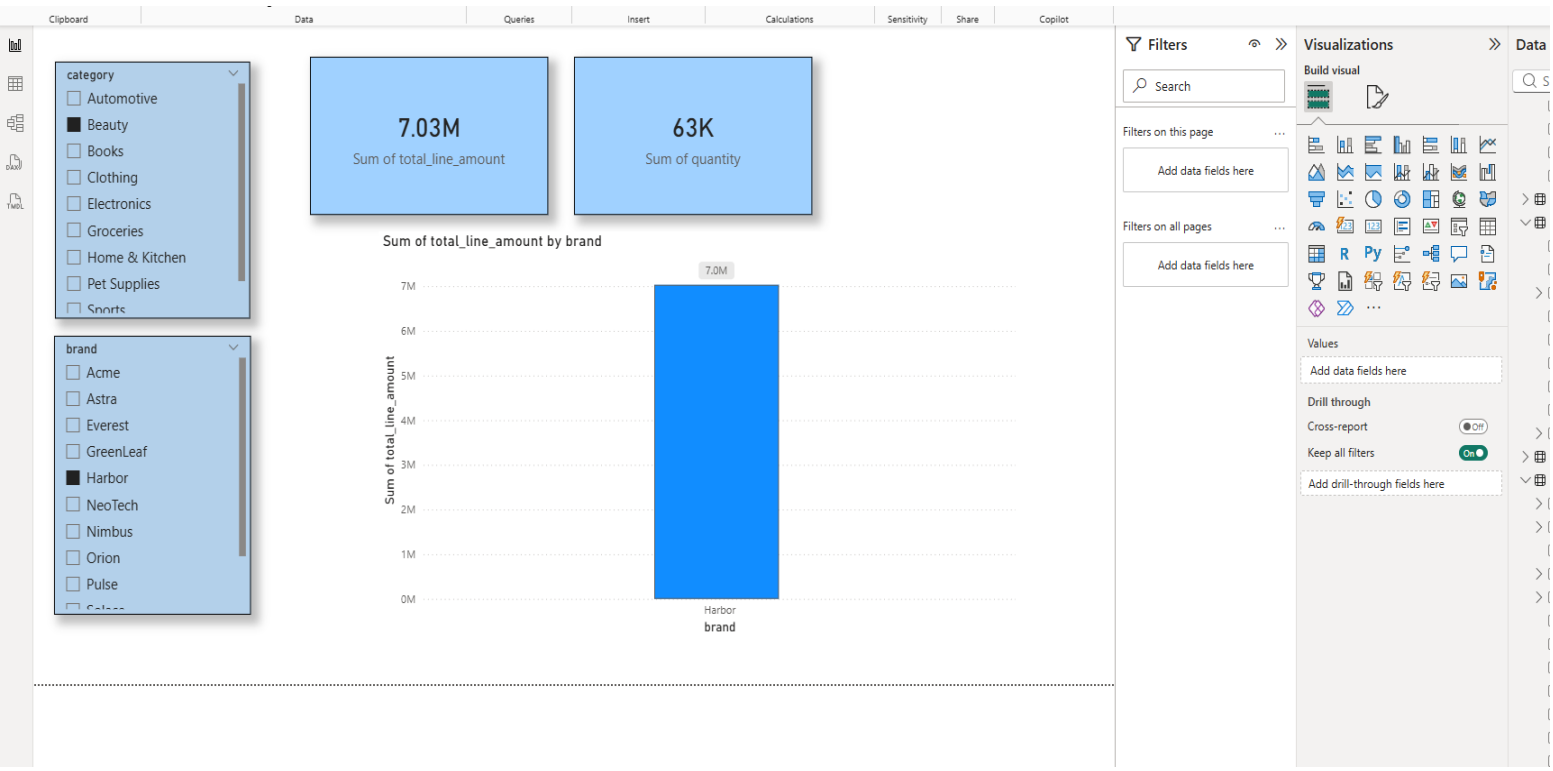
- Utilized Power BI's built-in aggregation functions to calculate metrics dynamically across different filter contexts.

- The interactive nature of this report allows stakeholders to perform multi-dimensional analysis by slicing data across time, product categories, and specific brands simultaneously.





d) Test the filters: Verify that selecting items in the slicers filters all visuals



5. Report 3: Drill-Down Report.

a) Built Hierarchical Visuals:

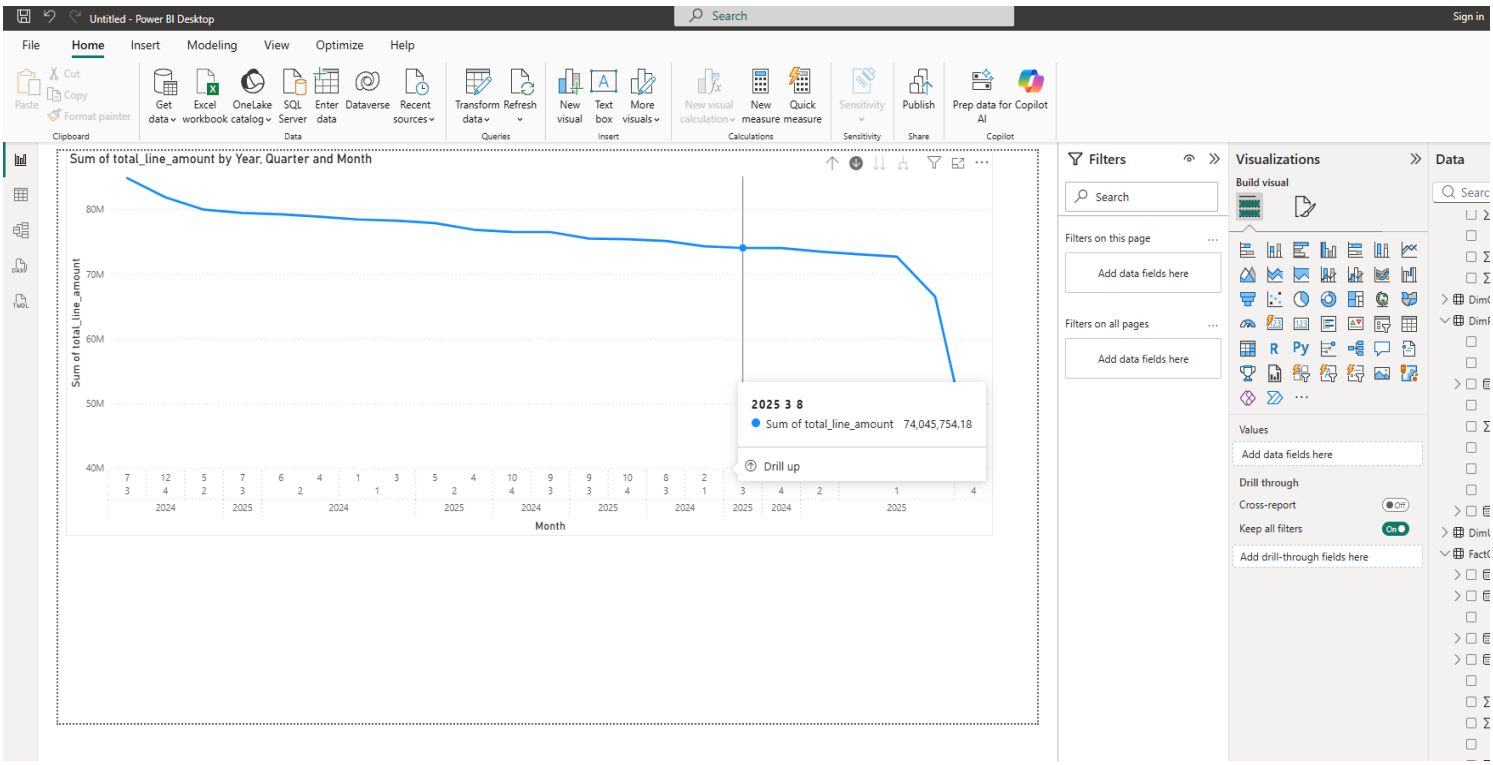
- Configured Column and Line Charts: Implemented a time-based hierarchy on the X-axis, starting from Year $\rightarrow$ Quarter $\rightarrow$ Month.
- Metric Selection: Used total_line_amount as the primary measure to analyze sales performance across these hierarchical levels.

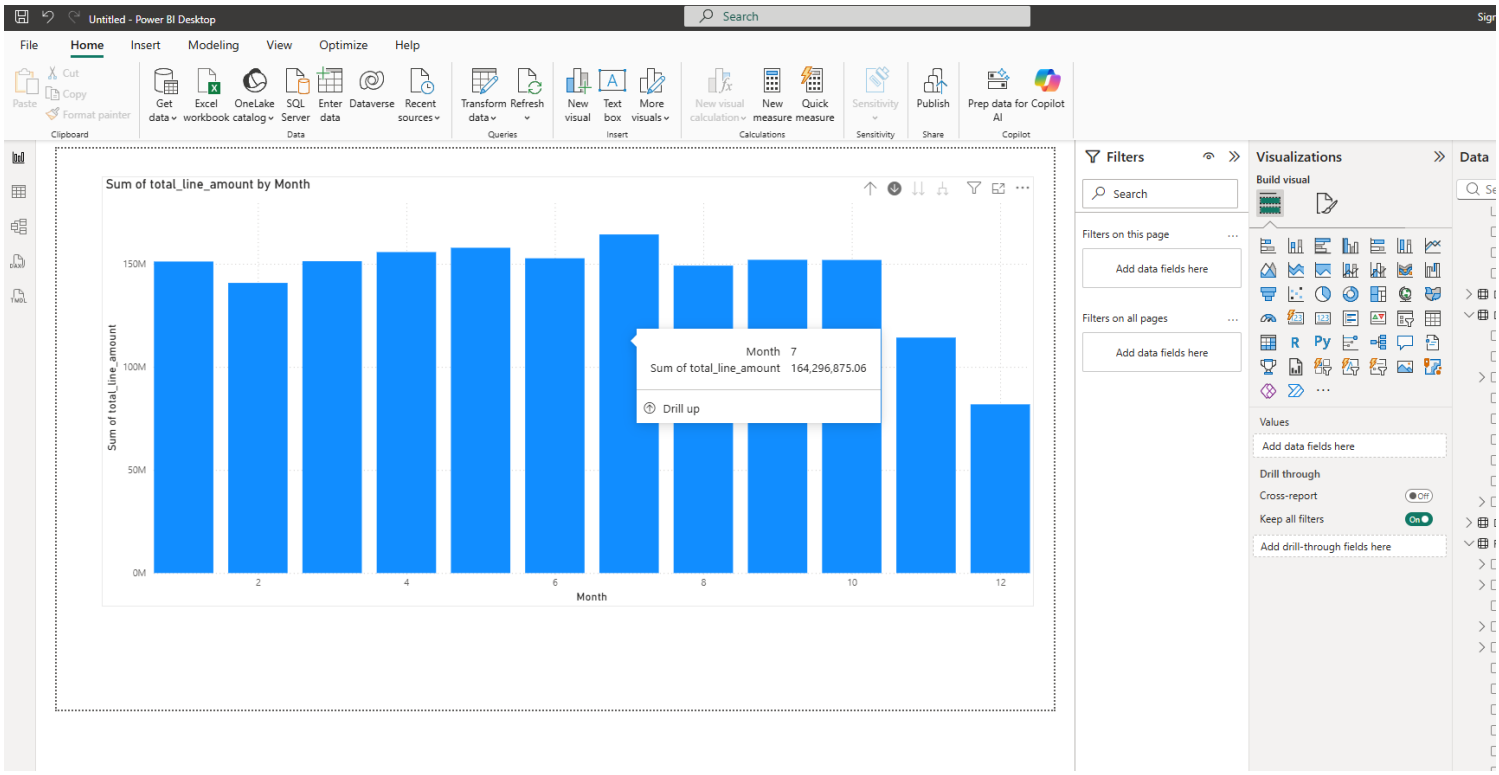
b) Drill-down Functionality:

- **Navigation:** Enabled drill-down features to allow users to explore data from a high-level annual summary down to specific monthly details.
- **Interactive Exploration:** Users can click on a specific year bar to view its constituent quarters, providing a deeper understanding of seasonal sales trends.
- **Visual Consistency:** Maintained the same hierarchy levels across both column and line charts to ensure a consistent analysis experience.

c) Data Insights:

- This hierarchical view helps in identifying specific months or quarters that contribute most to the overall annual revenue, allowing for more granular business decision-making.





6. Report 4: Drill-Through Report

a) Concept and Purpose:

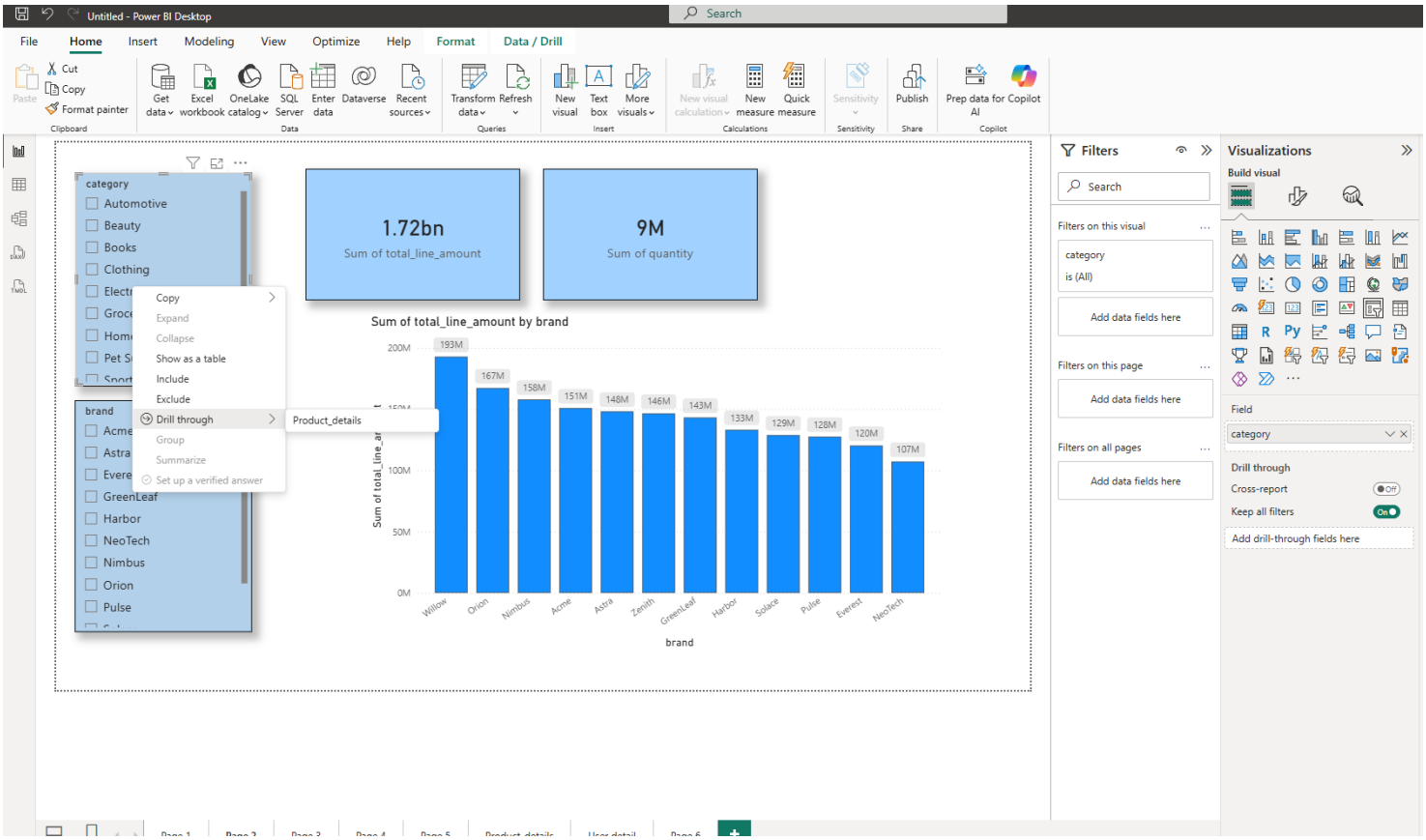
- The Drill-through feature was implemented to allow users to navigate from high-level summary charts to detailed transaction-level data.
- This provides a "deep dive" capability, where a user can analyze the specific products or users contributing to the totals seen on the main dashboard.

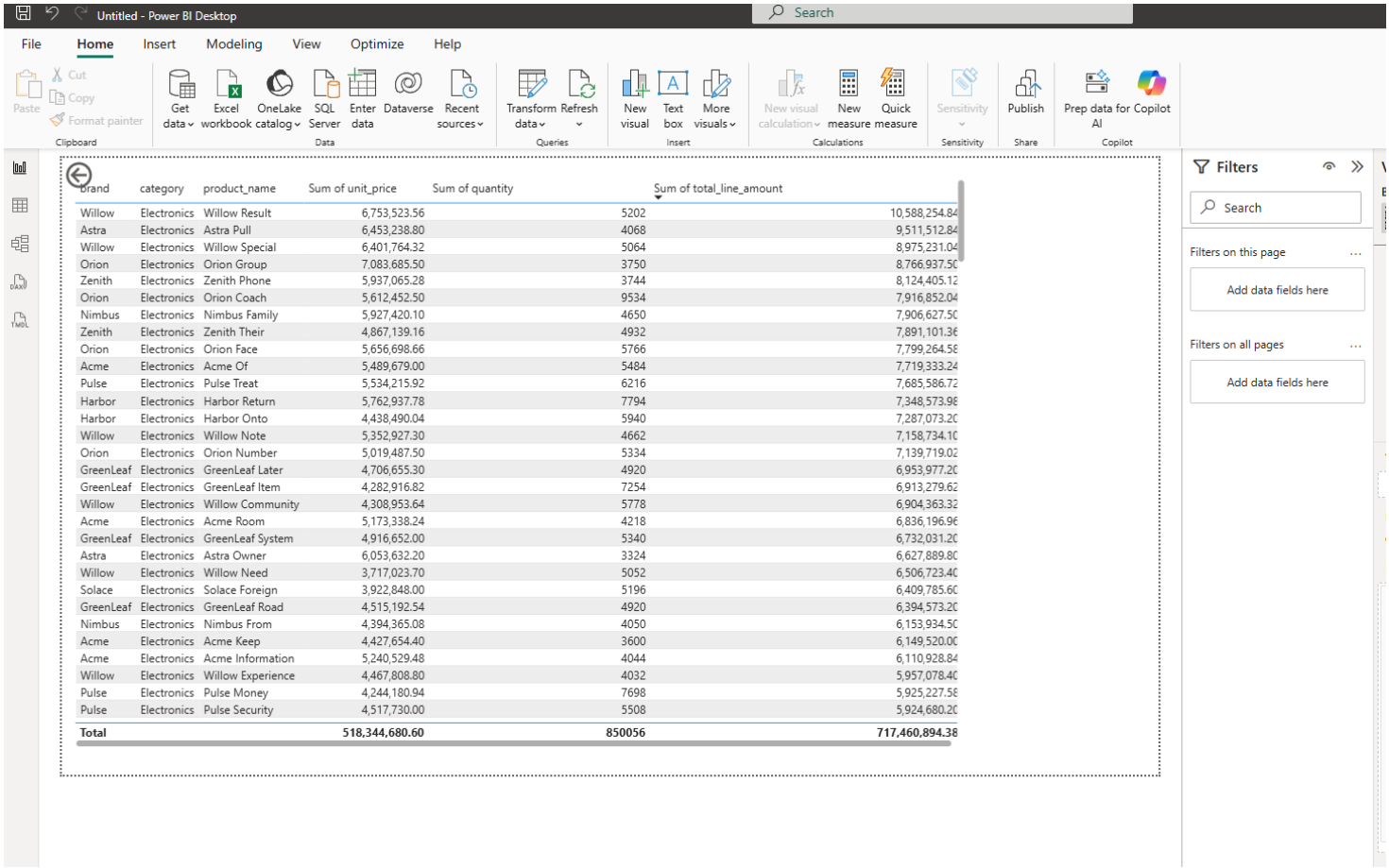
b) Implementation Steps:

1. Detail Page Setup: Created a dedicated page named "Product Detail" containing a table with specific attributes like Product Name, Brand, and Total Sales.
2. Defining the Drill-through Field: Added the category field from the DimProduct table into the Drill-through filter bucket on the detail page.
3. User Interaction: Configured the main "Sales by Category" bar chart to act as the source. When a user right-clicks a category bar, the "Drill-through" option becomes available.

c) Functional Testing:

- Action: Right-clicked on the 'Electronics' category in the main bar chart and selected Drill-through -> Product Detail.
- Result: The report successfully navigated to the detail page, automatically filtering the table to display only products belonging to the 'Electronics' category.
- Verification: Confirmed that the summary totals on the main page match the sum of detailed records on the drill-through page.





brand	category	product_name	Sum of unit_price	Sum of quantity	Sum of total_line_amount
Willow	Electronics	Willow Result	6,753,523.56	5202	10,588,254.84
Astra	Electronics	Astra Pull	6,453,238.80	4068	9,511,512.84
Willow	Electronics	Willow Special	6,401,764.32	5064	8,975,231.04
Orion	Electronics	Orion Group	7,083,685.50	3750	8,766,937.50
Zenith	Electronics	Zenith Phone	5,937,065.28	3744	8,124,405.12
Orion	Electronics	Orion Coach	5,612,452.50	9534	7,916,852.04
Nimbus	Electronics	Nimbus Family	5,927,420.10	4650	7,906,627.50
Zenith	Electronics	Zenith Their	4,867,139.16	4932	7,891,101.36
Orion	Electronics	Orion Face	5,656,698.66	5766	7,799,264.58
Acme	Electronics	Acme Of	5,489,679.00	5484	7,719,333.24
Pulse	Electronics	Pulse Treat	5,534,215.92	6216	7,685,586.72
Harbor	Electronics	Harbor Return	5,762,937.78	7794	7,348,573.98
Harbor	Electronics	Harbor Onto	4,438,490.04	5940	7,287,073.20
Willow	Electronics	Willow Note	5,352,927.30	4662	7,158,734.10
Orion	Electronics	Orion Number	5,019,487.50	5334	7,139,719.02
GreenLeaf	Electronics	GreenLeaf Later	4,706,655.30	4920	6,953,977.20
GreenLeaf	Electronics	GreenLeaf Item	4,282,916.82	7254	6,913,279.62
Willow	Electronics	Willow Community	4,308,953.64	5778	6,904,363.32
Acme	Electronics	Acme Room	5,173,338.24	4218	6,836,196.96
GreenLeaf	Electronics	GreenLeaf System	4,916,652.00	5340	6,732,031.20
Astra	Electronics	Astra Owner	6,053,632.20	3324	6,627,889.80
Willow	Electronics	Willow Need	3,717,023.70	5052	6,506,723.40
Solace	Electronics	Solace Foreign	3,922,848.00	5196	6,409,785.60
GreenLeaf	Electronics	GreenLeaf Road	4,515,192.54	4920	6,394,573.20
Nimbus	Electronics	Nimbus From	4,394,365.08	4050	6,153,934.50
Acme	Electronics	Acme Keep	4,427,654.40	3600	6,149,520.00
Acme	Electronics	Acme Information	5,240,529.48	4044	6,110,928.84
Willow	Electronics	Willow Experience	4,467,808.80	4032	5,957,078.40
Pulse	Electronics	Pulse Money	4,244,180.94	7698	5,925,227.58
Pulse	Electronics	Pulse Security	4,517,730.00	5508	5,924,680.20
Total			**518,344,680.60**	**850056**	**717,460,894.38**

Untitled - Power BI Desktop

brand	category	product_name	Sum of unit_price	Sum of quantity	Sum of total_line_amount
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GreenLeaf	Electronics	GreenLeaf Road	4,515,192.54	4920	6,394,573.20
Nimbus	Electronics	Nimbus From	4,394,365.08	4050	6,153,934.50
Acme	Electronics	Acme Keep	4,427,654.40	3600	6,149,520.00
Acme	Electronics	Acme Information	5,240,529.48	4044	6,110,928.84
Willow	Electronics	Willow Experience	4,467,808.80	4032	5,957,078.40
Pulse	Electronics	Pulse Money	4,244,180.94	7698	5,925,227.58
Pulse	Electronics	Pulse Security	4,517,730.00	5508	5,924,680.20
Total			**518,344,680.60**	**850056**	**717,460,894.38**

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

Values

Add data fields here

Drill through

Cross-report

Keep all filters

category
is Electronics
Allow drill through when:
Used as category

Beauty	207
Books	194
Clothing	213
Electronics	203
Groceries	183
Home & Kitchen	199
Pet Supplies	212
Sports	183
Toys	213